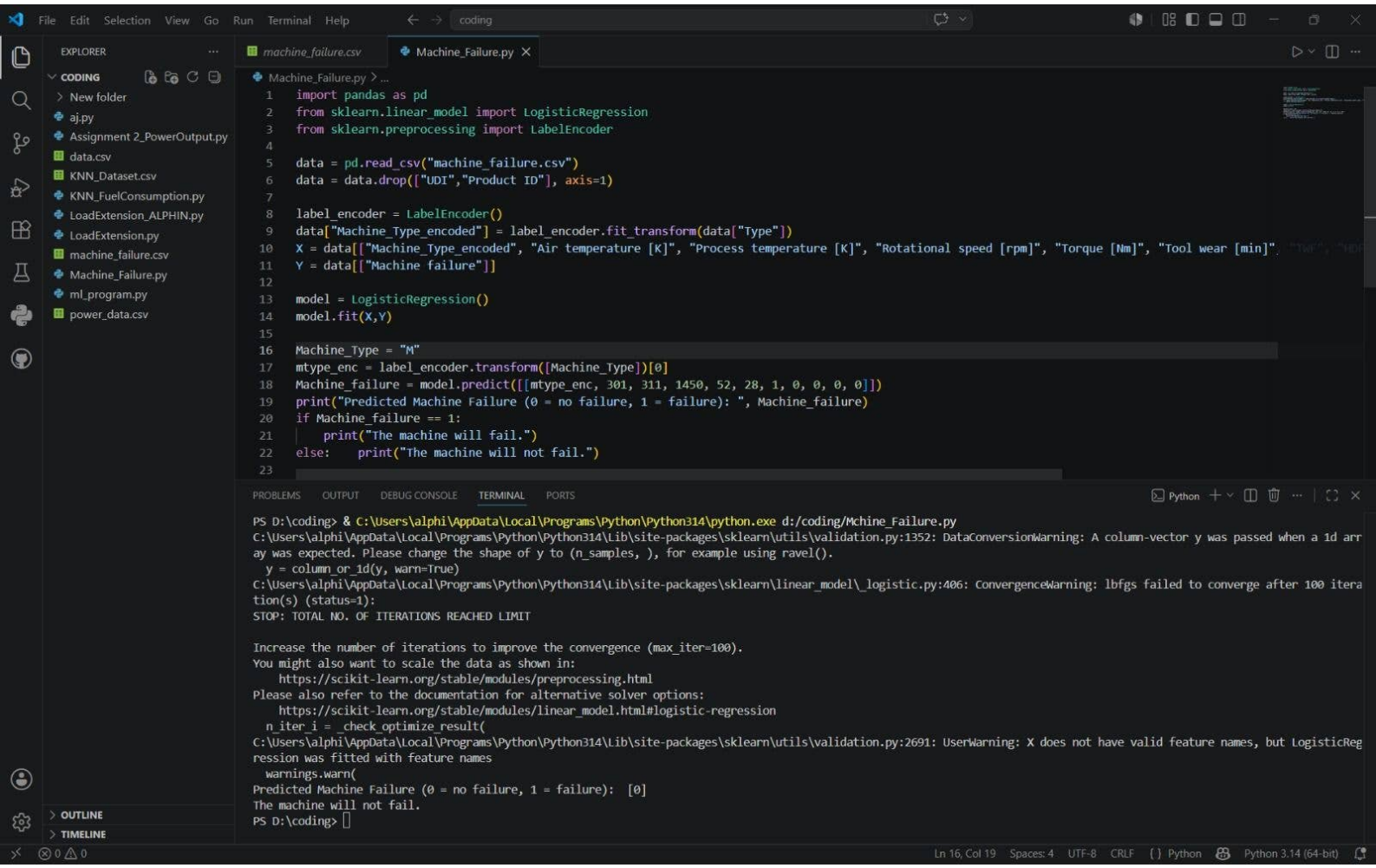
File Edit Selection View Go Run Terminal Help
coding
EXPLORER
CODING
New folder
aj.py
Assignment 2_PowerOutput.py
data.csv
KNN_Dataset.csv
KNN_FuelConsumption.py
LoadExtension_ALPHIN.py
LoadExtension.py
machine_failure.csv
Machine_Failure.py
ml_program.py
power_data.csv
machine_failure.csv
Machine_Failure.py
Machine_Failure.py > ...
1 import pandas as pd
2 from sklearn.linear_model import LogisticRegression
3 from sklearn.preprocessing import LabelEncoder
4
5 data = pd.read_csv("machine_failure.csv")
6 data = data.drop(["UDI","Product ID"], axis=1)
7
8 label_encoder = LabelEncoder()
9 data["Machine_Type_encoded"] = label_encoder.fit_transform(data["Type"])
10 X = data[["Machine_Type_encoded", "Air temperature [K]", "Process temperature [K]", "Rotational speed [rpm]", "Torque [Nm]", "Tool wear [min]"]]
11 Y = data[["Machine failure"]]
12
13 model = LogisticRegression()
14 model.fit(X,Y)
15
16 Machine_Type = "M"
17 mtype_enc = label_encoder.transform([Machine_Type])[0]
18 Machine_failure = model.predict([[mtype_enc, 301, 311, 1450, 52, 28, 1, 0, 0, 0, 0]])
19 print("Predicted Machine Failure (0 = no failure, 1 = failure): ", Machine_failure)
20 if Machine_failure == 1:
21 print("The machine will fail.")
22 else: print("The machine will not fail.")
23
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
PS D:\coding> & C:\Users\alphi\AppData\Local\Programs\Python\Python314\python.exe d:/coding/Mchine_Failure.py
C:\Users\alphi\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py:1352: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples, ), for example using ravel().
y = column_or_1d(y, warn=True)
C:\Users\alphi\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\linear_model_logistic.py:406: ConvergenceWarning: lbfgs failed to converge after 100 iteration(s) (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT
Increase the number of iterations to improve the convergence (max_iter=100).
You might also want to scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
C:\Users\alphi\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
warnings.warn(
Predicted Machine Failure (0 = no failure, 1 = failure): [0]
The machine will not fail.
PS D:\coding>
OUTLINE
TIMELINE
Ln 16, Col 19 Spaces: 4 UTF-8 CRLF { } Python Python 3.14 (64-bit)